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Water pollution related to food consumption patterns:

Dietary choices cause grey water footprints due to agricultural pesticide use, a case study for China

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Abstract

Food production and consumption affect environmental sustainability, also causing water pollution. This study assesses the relationship between food consumption patterns and water pollution using China as the case study, with a specific emphasis on pesticide-related pollution using the grey WF as the indicator. It includes the whole production chain, from fork to farm, considering food waste and losses. It assesses grey WFs for the current Chinese food consumption pattern, the recommended pattern of the Chinese government, the 'planetary health diet' and a plant-based diet. The current pattern, high in animal foods, has the largest grey WF (42 m³/day), despite animal foods providing less energy compared to crop-based foods. Pork, beef and potato, only 8% of energy consumption, contribute 73% to total grey WFs, indicating the highest potential water pollution impact. The recommended pattern has a grey WF of 26, the planetary health diet of 21, and the plant-based pattern of 11 m³/day. The temporal grey WFs peak during the cropping season. The current food consumption pattern has the highest peak, the plant-based pattern the lowest. Especially animal foods substantially increase grey

WFs in summer due to feed production (e.g. maize). Therefore, the current and recommended diets with relatively much animal foods have a larger impact on grey WF peaks in the cropping season than the planetary health and plant-based patterns. Transitioning to more sustainable food consumption patterns with less animal foods in combination with better pesticide management, especially for maize and potatoes, reduces water pollution. Our results comply to results of earlier studies indicating the environmental impact of patterns high in animal foods. Although the study has taken China as the case study area, the methodological framework is also relevant on a global scale, where food consumption choices affect environmental sustainability, including water pollution, provided that country-specific data are applied.

Keywords: Food consumption patterns, Water pollution, Grey water footprint, Pesticides, China, Diets

1 Introduction

Globally, food production and consumption affect environmental sustainability, also causing water pollution (FAO, 2012). Water pollution refers to the introduction of harmful substances, such as nutrients, pesticides, and other chemicals, into freshwater systems, degrading water quality and threatening human and ecological health. Freshwater of sufficient quality is essential for human health and well-being, but anthropogenic water pollution, such as contamination by pesticides applied in agriculture, significantly degrades water quality and poses risks to both aquatic ecosystems and human health (Schwarzenbach et al., 2010; Wang et al., 2024). While previous studies have extensively examined GHG emissions (e.g., Crippa et al., 2021), land use (e.g., Kastner et al., 2012), and nutrient-related water pollution (e.g., Mekonnen & Hoekstra, 2015, 2018), the role of pesticides in water pollution from food systems remains underexplored. The Chinese Ministry of Agriculture, for example, agreed on the strategy ‘Zero Growth of Pesticide till 2020’ (MOA, 2015).

With the increasing global population, food security issues and food production within the planetary boundaries have become important concerns (e.g., Foley et al., 2011; Godfray et al., 2010; Tilman et

al., 2011). Food production uses natural resources (e.g., land, water, nitrogen and phosphorus) and causes emissions to the environment, e.g. greenhouse gas (GHG) emissions or agricultural emissions to water bodies (e.g., nitrogen, phosphorus or pesticides). However, different food commodities affect environmental sustainability in a different way, depending on their production system. This means that there are huge differences among the environmental impacts of food consumption patterns depending on their composition. Many studies have shown that especially consumption patterns with a large contribution of animal foods have larger impacts than plant-based patterns. Indicators applied include the ecological footprint, land requirements, GHG emissions or water footprint, among others. The ecological footprint developed by Wackernagel and Rees (1998) addresses the need for biophysical approaches to achieve environmental sustainability indicating the large land use especially related to food consumption. Gerbens-Leenes and Nonhebel (2002) have shown that different food consumption patterns have different agricultural land requirements, while dietary changes are the predominant driver of future land needs for food (Kastner et al., 2012). The intensification of food systems and shifts of human diets to more animal foods have caused an increase of GHG emissions. For example, Steinfeld et al. (2006) revealed that animal production is responsible for approximately 18% of global GHG emissions. Recently, Crippa et al. (2021) have reported that food systems contribute 34% to global GHG emissions. Also water use and water pollution related to food consumption and production are relevant for environmental sustainability. For instance, Hoekstra and Mekonnen (2012) quantified the green, blue and grey water footprints (WFs) of humanity, including water for food. Mekonnen and Hoekstra (2015, 2018,) assessed global water pollution caused by nitrogen and phosphorus, fertilizers used for food production. For the assessment, they applied the grey WF, defined as the hypothetical volume of freshwater required to assimilate a pollutant load to the extent that ambient water quality standards are not violated. It serves as a volumetric indicator of water pollution, not as the actual volume of water polluted or required for dilution (Hoekstra et al., 2011). Franke et al. (2013) updated the grey WF methodology, also taking, among other things, pesticides into consideration.

To assess water pollution, the methods of the grey water footprint (WF) and life cycle assessment (LCA) are useful and complementary tools. The grey WF is able to measure total pollution levels, whereas LCA supports comprehensive sustainability assessments (Berger et al., 2025). Where the grey

WF aims to indicate volumetric water needs required to assimilate water pollution by the ambient water system, LCA's main goal is to quantify potential environmental impacts resulting from the emission of pollutants, including eutrophication, acidification, human health and ecotoxicity (Berger et al., 2025). In this study, we aimed to estimate the impact of different food consumption patterns on water pollution and therefore selected the concept of the grey WF for the analysis.

Recently, Yi et al. (2024a) have assessed grey WFs related to pesticides for the main crops produced in China, indicating not only that the current crop grey WFs are dominated by pesticides, and not by nitrogen or phosphorus, but also that there are huge differences among crops. Moreover, animal production systems convert large quantities of feed crops into animal foods (meat, milk and eggs), so that their water pollution is probably larger than pollution caused by vegetal foods (e.g. grains, vegetables and fruits). This means that differences in the composition of food consumption patterns might have different effects on water pollution. At present, there is no quantification of water pollution caused by food consumption patterns. To achieve a more sustainable food consumption pattern, addressing environmental and health concerns, Duchin (2005), for example, proposed a predominantly plant-based Mediterranean-type pattern. Willett et al. (2019) proposed the 'planetary health diet', a food consumption pattern with a small contribution of animal foods and relatively small environmental footprints, including GHGs, water, land, N and P use, and biodiversity loss. An important question would be whether or not food consumption patterns dominated by animal foods would also cause more water pollution than patterns mostly based on vegetal foods.

Food consumption patterns in China have shown large changes in the recent past, especially the consumption of animal foods went up (FAO, 2024). The Chinese government recommends healthy food consumption patterns including milk and meat (Wang et al., 2016). Importantly, this study adopts a national systems perspective rather than focusing on regional heterogeneity within China. While regional dietary variations exist, the central government's unified food and nutrition guidelines provide a coherent national policy framework. This study aims to find the relationship between food consumption patterns and water pollution using China as the case study area. It adopts the grey WF (Franke et al., 2013) to assess water pollution for four different food consumption patterns, the current Chinese food consumption pattern (averaged on the national level for the year 2020), the recommended

pattern of the Chinese government, the ‘planetary health diet’ and a plant-based diet in which all animal based foods are replaced by plant based foods. The research questions are: 1) Which pollutants dominate the grey WFs for food commodities at the farm gate?; 2) What are the grey WFs per unit of food commodity or per unit of food at the farm gate, retail gate and in the consumer phase?; 3) What are the grey WFs of the current Chinese food consumption pattern, the recommended pattern of the Chinese government, the ‘planetary health diet’ and of a plant-based diet?

The study gives insight into water pollution related to dietary choices. This information is not only relevant for China, but also for other countries.

2 Grey WFs related to food supply chains

Food commodities produced in agriculture show losses in the supply chain from the farm gate to final consumption. This study employs a consumption-based accounting framework to assess water pollution associated with food consumption. Unlike production-based assessments, which allocate environmental impacts solely to the point of emission, a consumption-based perspective traces impacts back to the final consumer, thereby revealing how downstream dietary choices drive upstream agricultural pollution. Assessments of environmental sustainability of food consumption should include the gap between food consumption and production. For example, Song et al. (2015) quantified the embedded GHG and water use caused by food waste in households and Xue et al. (2021) assessed the food losses and waste for the whole food supply chain, from the farm gate, the retail gate, to final consumption (eating of the food, e.g. a strawberry), including the main crops and animal and aquatic foods in China. That study has shown that different foods have different losses in their supply chain. Supply chain losses cause an increase of grey WFs per unit of product. When food is lost or wasted, the environmental burdens embedded in the lost portion—including pesticide-related water pollution—are effectively allocated to the actual food consumed, amplifying the per-unit environmental footprint. Figure 1 shows an example of a grey water footprint (WF) increase for strawberries due to food losses along a food supply chain.

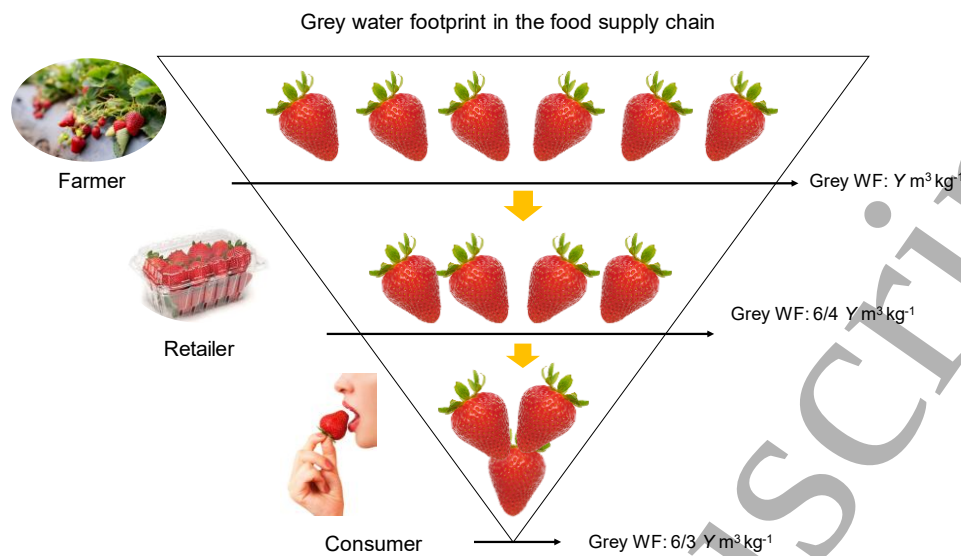


Figure 1 Grey water footprint (WF) increase due to food losses along a food supply chain from farm gate, to retail gate, to consumer final consumption

Figure 1 illustrates the grey WF increase per unit of food as it progresses from farmers to retailers and finally to consumers. This amplification effect has critical policy implications: when consumers choose foods with high supply chain losses, they indirectly cause a larger environmental impact than reflected in farm-gate assessments. Using strawberries as an example: farmers produce six strawberries (at the farm gate), with a grey WF of $Y \text{ m}^3$ per kg of strawberry. As the strawberries move through the supply chain, at the retail gate, there are only four left, causing the grey WF to increase to $6/4 Y \text{ m}^3$ per kg. By the time the strawberries reach the consumer's mouth, only three strawberries remain, resulting in a grey WF of $6/3 Y \text{ m}^3$ per kg. Overall, the stages of food production, supply, and consumption significantly impact the grey WF. Additionally, consumer choices regarding different foods with different initial grey WFs and different losses will result in varying grey WFs.

2.1 Farmers and food production

Farmers produce food commodities, crop based, animal based and aquatic commodities for human consumption. Yi et al. (2024a) quantified grey WFs related to pesticides, phosphorus and nitrogen for

crop production in China, showing that crop grey WFs are dominated by pesticides. For animal and aquatic production, grey WFs not only depend on pesticide use for feed crops, but also the amount of feed ingredients per animal (Lu et al., 2024; Wang et al., 2007; Yi et al., 2023) or fish intake (Jiang et al., 2022). As a result, the size of grey WFs per unit of commodity output at the farm gate shows large differences.

2.2 Retailers and food supply

After leaving the farm gate, food commodities often undergo processing and distribution to retailers. During this process, food conversions and food waste and loss (FWL) occur. In China, different food commodities have different FWLs. For example, fresh strawberries have a FWL of around 33% from farm gate to retail gate, while the FWL for cereals is about 10%, and for vegetables around 50% (Xue et al., 2021). However, only 45% of the live-weight of beef cows is converted into beef after slaughter, with an average FWL for Chinese beef of around 20% at the retail gate (FAO, 2003; Xue et al., 2021).

2.3 Consumers and food consumption

Finally, consumers buy the foods at the retail. There are also conversions, wastes and losses in the consumer stage. These conversions vary among different foods. Some foods, like rice, wheat and soybean based foods (e.g., boiled rice, noodles and home-made soy milk) increase their weight, because they take up water. Other foods, like beef, lose weight when cooked or by peeling, like potato or orange. Additionally, between the food at the retail gate and the food actually consumed, there is also FWL.

Consumers face thousands of food choices for their food consumption pattern. These patterns have different grey WFs, depending on the composition.

2.4 System description and boundary

The system included in this study covers the full food supply chain from agricultural production (farm) through processing and distribution (food industry and retail) to final consumption (fork). The assessment includes: (1) direct pesticide, nitrogen, and phosphorus emissions from crop production; (2) indirect emissions embedded in feed crops used for animal and aquatic production; and (3) amplification effects from supply chain losses and conversions. The following two aspects that are not taken into account: (1) Regional heterogeneity: While China exhibits regional variations in dietary patterns and agricultural practices, this study adopts a national aggregate perspective. This approach aligns with the central government's unified food and nutrition policy framework; (2) Agricultural international trade: China's staple foods (rice, wheat and potato), fruits and vegetables, animal feed (e.g. maize) and livestock foods (pork, poultry, beef, milk) are largely produced in China itself (FAO, 2024). The study focuses on domestic consumption-production linkages, and trade-related impacts in other countries are not considered. This exclusion is partially justified by China's national food security strategy, which prioritizes domestic production for staple crops. However, we acknowledge that China is a significant importer of soybeans and other feed crops, meaning that a portion of the grey WF embodied in Chinese consumption may occur outside China's borders and is not captured in this analysis.

3 Method and data

For the assessment of grey WFs of food consumption patterns, this study selected four patterns: 1) the current Chinese pattern based on food supply in 2020 (FAO, 2024); 2) the recommended pattern based on China's governmental food recommendations (Wang et al., 2016), 3) the planetary health diet (Willett et al., 2019) and 4) a plant-based diet where animal foods of pattern 2 are replaced by soybean and other legumes products. Pattern 2, 3 and 4 are scenario patterns adjusted to a recommended nutritional energy intake of 2505 Kcal per capita per day, allowing for comparisons of dietary composition while controlling for total energy. The food consumption pattern 1-3 include 22 food items: 7 animal foods (beef, beef (milking cow), pork, poultry, sheep meat, milk and eggs), 14 crop-based

foods (rice, wheat, maize, soybean, soybean milk, peanut (nuts), potato, vegetables, fruits, soybean oil, peanut oil, rapeseed oil, sugarcane and sugar-beet) and one aquatic food (fish).

To assess the grey WF of the four food consumption patterns, this study used four calculation steps: the calculation of 1) grey WFs of food commodities at the farm gate; 2) grey WFs of foods at the retail gate; 3) grey WFs for foods by consumers and 4) grey WFs of the four food consumption patterns. Figure 2 shows the calculation steps and data inputs.

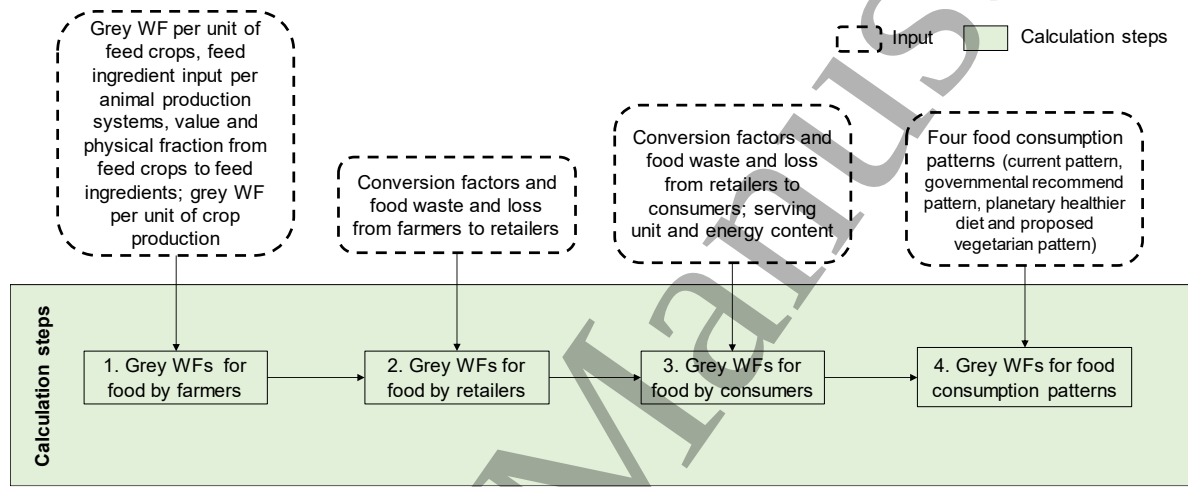


Figure 2 Calculation steps for the assessment of grey WFs for food consumption patterns in China along the supply chain, from farmers to retailers and to consumers

3.1 Step 1: calculation grey WFs for food commodities by farmers

Step 1 calculates the grey WF for food commodity f per pollutant p at time t at the farm gate, *Grey* $WF_{f,p,t,farmers}$ ($m^3 kg^{-1}$), including crop, animal and aquatic production. For animal and aquatic production, the *Grey* $WF_{f,p,t,farmers}$ was calculated as:

$$Grey WF_{f,p,t,farmers} = \frac{\sum_j (Q_{i,j} \times Grey WF_{j,p,t} \times (VF_j \div PF_j))}{1000} \quad (1)$$

where Q_{ij} is the amount of feed ingredient j for animal and aquatic production system i (kg dry matter per unit of animal product). Grey $WF_{j,p,t}$ is the grey WF per unit of feed crop j ($\text{m}^3 \text{ton}^{-1}$) per pollutant p at time t . The time t has two scales: 1) the year of pesticide and fertilizer applied; 2) a specific day within a year from 0 to 365 (i.e. from January 1 to December 31). VF_j and PF_j are the value fraction and product fraction of feed ingredient j from feed crop j . The value fraction (VF_j) is used for economic allocation, determining the share of the total grey WF allocated to each co-product from the multi-output processing of feed crop j . The product fraction (PF_j) converts the crop-level grey WF to the specific feed ingredient level, enabling the calculation of grey WF intensity per unit of feed ingredient. These two fractions serve different purposes, allocation and mass conversion, and their product does not constitute double-counting. This allocation approach follows the methodology established in Hoekstra et al. (2011). The factor 1000 converts tons to kg. For aquatic production, the analysis includes carp species (*Cyprinus carpio*), which account for approximately 70% of China's aquaculture production (FAO, 2024). This simplification may introduce a bias, likely overestimating the grey WF for the aquatic food category, as carp production requires formulated feed while capture fisheries do not. Feed composition data for carp aquaculture are derived from Jiang et al. (2022). Table s1 in the supporting information (SI) gives data and references on Q_{ij} . For crop production, we derived data on crop grey WFs from Yi et al. (2024ab). Table s2 and s3 in the SI give the grey WFs and references. Table s4 in the SI data and references on VF_j and PF_j .

The pesticide-related grey WF calculations in this study follow the methodology established by Franke et al. (2013) and applied to China by Yi et al. (2024a, 2024b). Pesticides considered include major categories applied to Chinese crops: herbicides (e.g., atrazine, glyphosate), insecticides (e.g., chlorpyrifos, imidacloprid), and fungicides (e.g., tebuconazole, pyraclostrobin). Data on pesticide application rates, application timing, and crop-specific usage patterns are derived from provincial agricultural statistical yearbooks and validated against national pesticide usage surveys (Yi et al., 2024a).

A critical methodological consideration is that the grey WF approach assumes pollutants behave conservatively—i.e., they do not degrade or adsorb in soil before reaching water bodies. In reality, pesticide environmental behavior varies considerably: some pesticides are moderately mobile and

degrade relatively quickly, while others are persistent and highly mobile (Franke et al., 2013). By using the traditional, conservative assumption proposed by Hoekstra et al. (2011) and Franke et al. (2013) our grey WF estimates represent a worst-case scenario. This methodological choice is consistent with the grey WF accounting guidelines (Franke et al., 2013) and ensures comparability with nutrient-related grey WFs, which also employ conservative assumptions for nitrogen and phosphorus.

3.2 Step 2: calculation grey WFs for food by retailers

Step 2 calculates the grey WF for foods at the retail gate. The grey WF per unit of food product r per pollutant p at time t , $Grey WF_{r,p,t,retailers}$ ($m^3 kg^{-1}$), was calculated as:

$$Grey WF_{r,p,t,retailers} = \frac{Grey WF_{f,p,t,farmers}}{CF_{f,r} * (1 - FWL_{f,r})} \quad (2)$$

where $CF_{f,r}$ and $FWL_{f,r}$ are the conversion factor and the food waste and loss rate for food from the farm gate to food r at the retail gate. Table s5 in the SI gives data and references on $CF_{f,r}$ and $FWL_{f,r}$.

3.3 Step 3: calculation grey WFs for food by consumers

Step 3 calculates the grey WF for the food consumed in consumer's mouth (m^3 per kg). The grey WF per unit of food c consumed per pollutant p at time t , $Grey WF_{c,p,t,consumers}$ ($m^3 kg^{-1}$), was calculated as:

$$Grey WF_{c,p,t,consumers} = \frac{Grey WF_{f,p,t,retailers}}{CF_{r,c} * (1 - FWL_c)} \quad (3)$$

where $CF_{r,c}$ is the conversion factor for food r from the retail gate to food c consumed by the consumer and FWL_c is the food waste and loss rate by consumers from retailer gate to the mouth. Table s5 in the SI gives data and references on $CF_{r,c}$ and FWL_c . Results were also expressed as grey WFs per serving

unit and grey WF per unit of nutritional energy (Kcal) Table s5 in the SI also gives data and references on serving unit and calorie content per food commodity.

3.4 Step 4: calculation grey WFs of food consumption patterns

Step 4 calculated the grey WFs for the four food consumption patterns. The grey WF for pattern D including food C related to pollutant p at time t , $Grey WF_{pattern_D, p, t}$ ($m^3 day^{-1}$), was calculated as:

$$Grey WF_{pattern_D, p, t} = \sum_C \frac{Q_{D,C} * Grey WF_{C, p, t, consumers}}{1000} \quad (4)$$

where $Q_{D,C}$ is the amount of food C ($g day^{-1}$) in daily food consumption pattern D . Table s6 in the SI gives data and references on $Q_{D,C}$. The factor 1000 converts grams (g) into kilograms (kg).

4 Results

This section presents the grey WFs associated with food production and consumption in China. First, it gives the grey WFs linked to different pollutants (nitrogen, phosphorus, and pesticides) across various crops and animal foods (Section 4.1). Next, it shows the grey WFs along the food supply chain, from farm-gate to final consumption, indicating the increase of grey WFs along supply chains due to food waste and losses (Section 4.2). Finally, it gives grey WFs of four food consumption patterns showing the impact of food choices on water pollution (Section 4.3).

4.1 Grey WFs for food production related to different pollutants

Figure 3 shows the grey WFs related to pesticides, phosphorus and nitrogen for animal, aquatic and crop based food commodities at the farm gate.

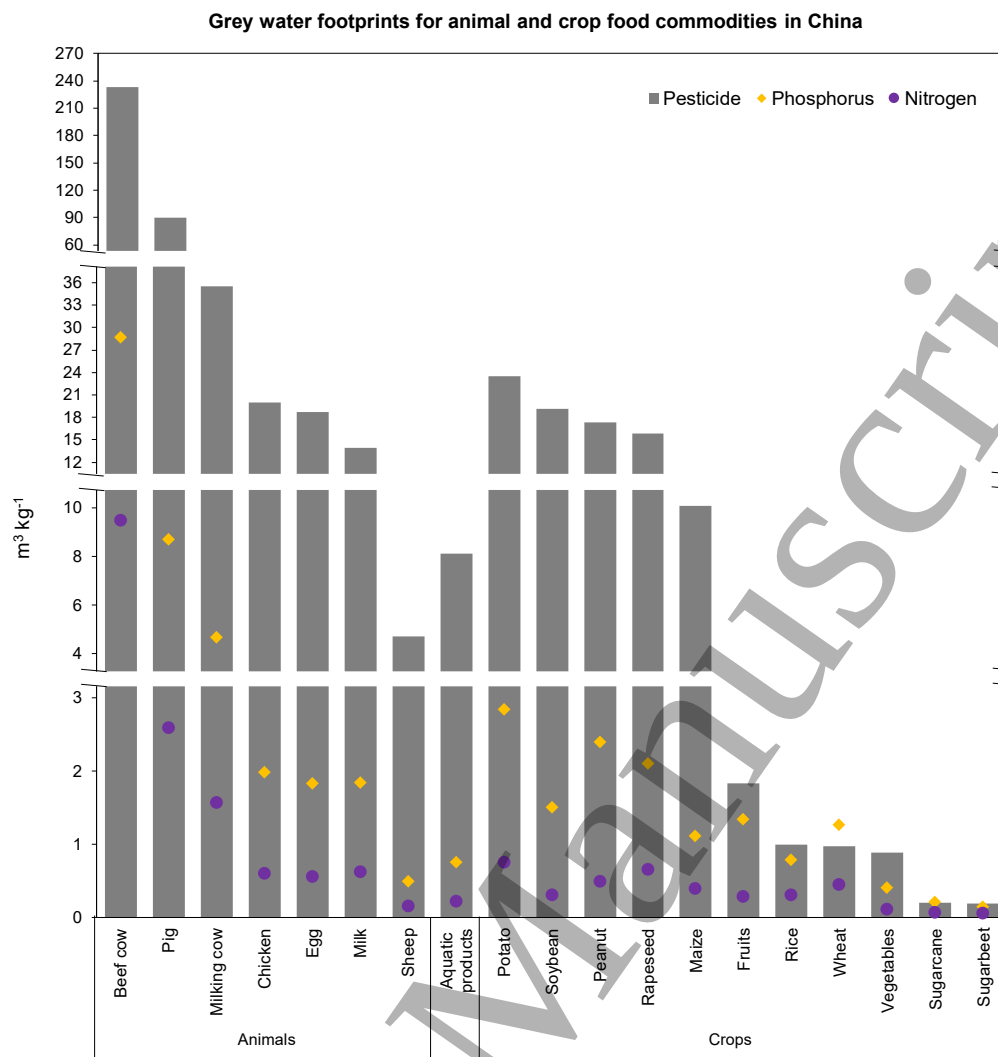


Figure 3 Grey water footprints related to pesticides, phosphorus and nitrogen for animal, aquatic and crop based food commodities at the farm gate in China

Figure 3 illustrates that pesticides dominate the grey water footprints (WFs) for all animal, aquatic and crop-based food commodities at the farm gate, except for wheat and sugarcane, where grey WF's due to phosphorus are largest. The differences between grey WF's related to pesticides and phosphorus range from 0.8 for wheat to 12.7 for soybeans. Similarly, the differences between pesticides and nitrogen range from 2.2 for wheat to 63 for soybeans.

Animal commodities exhibit larger grey WF's compared to crop-based commodities. For instance, the grey WF for beef cow is the largest, at 230 m^3 per kg of live weight and 647 m^3 per kg of carcass weight (Figure 4), which is nearly nine times larger than the grey WF of potatoes when compared at the farm

gate. Within the same food categories, significant variations in grey WFs are observed. For example, grey WFs for animal commodities range from 4.7 m³ per kg of live weight for sheep to 230 m³ per kg of live weight for beef cows. Similarly, among crop-based commodities, the grey WF of sugarbeet is smallest with 0.18 m³ per kg, which is only 1% of the grey WF of potatoes.

4.2 Grey WFs for food from the farm-gate to final consumption

Figure 4 shows the grey WF for animal, aquatic and crop food commodities at the farm gate, foods at the retail gate, and final consumption (serving unit).

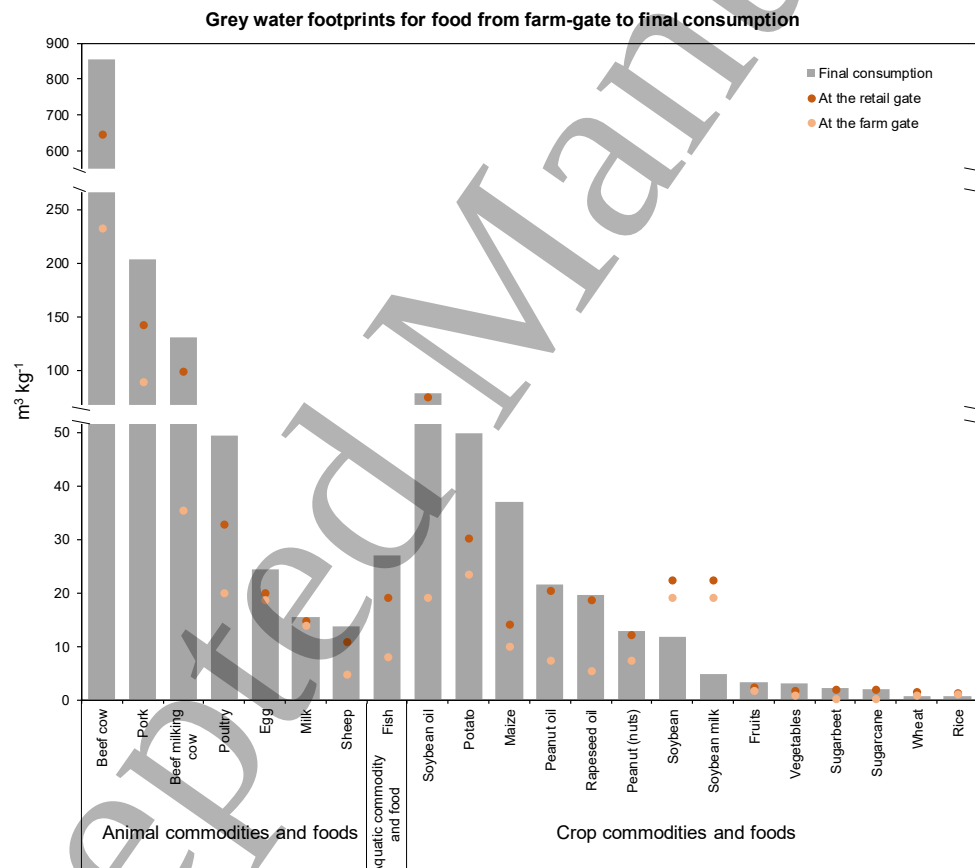


Figure 4 Grey WFs of animal, aquatic and crop food commodities at the farm gate, foods at the retail gate, and foods for final consumption

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Supply chain losses amplify grey WFs for most foods, with the magnitude of amplification varying considerably by food type. Figure 4 shows that in the food supply chain the grey WFs for most foods increase gradually from the production stage at the farm gate to the retailers and finally to the consumer in the mouth, excluding foods derived from rice and wheat. For instance, the grey WFs for potatoes and vegetables are 23 and 0.9 m³ per kg at the farm gate, rising to 30 and 1.8 m³ per kg at the retail gate, and finally to 50 and 2.9 m³ per kg when finally consumed. For potatoes, this represents more than a two-fold amplification. For rice and wheat, however, the grey WFs increase from farmers to retailers due to conversion and food waste and loss. From retailers to consumers, the grey WFs per kg of food decrease because converting raw commodities to cooked food on the table increases the food weight due to water uptake. This does not represent a real reduction in pollution, but reflects the change in the per-kilogram normalization basis. For animal foods and oil-crop-based foods, the conversion from animals and crops to products results in larger grey WFs. This amplification effect suggests that reducing food loss and waste, particularly for high-grey-WF foods like potatoes, can be an effective strategy to lower the grey WF of consumption.

The grey WF per serving unit varies for different foods. Comparing potatoes and rice, the grey WF for rice is smaller. For beef, including beef from milking cows, and sheep, the grey WF for sheep is smaller. In terms of energy intake, Table s7 in the SI shows that the grey WFs per 1,000 kcal intake are smallest for rice and wheat and largest for potatoes among crop-based foods. For animal-based foods, the grey WF per unit of energy intake is largest for beef and smallest for sheep meat. For food consumed, replacing, for example, a serving unit of milk by soybean milk reduces the grey WF from 2.2 m³ to 0.67 m³. The grey WF for beef produced by milking cows is smaller than for beef from beef cows (Table s7 in the SI).

4.3 Grey WFs and temporal variation of grey WFs of food consumption patterns

Figure 5 shows the food amounts (5a), energy intakes (5b) and grey WFs (5c) for the four Chinese food consumption patterns: the current pattern based on China's food supply in 2020, the recommended governmental pattern, the planetary health diet and the proposed plant-based food consumption pattern.

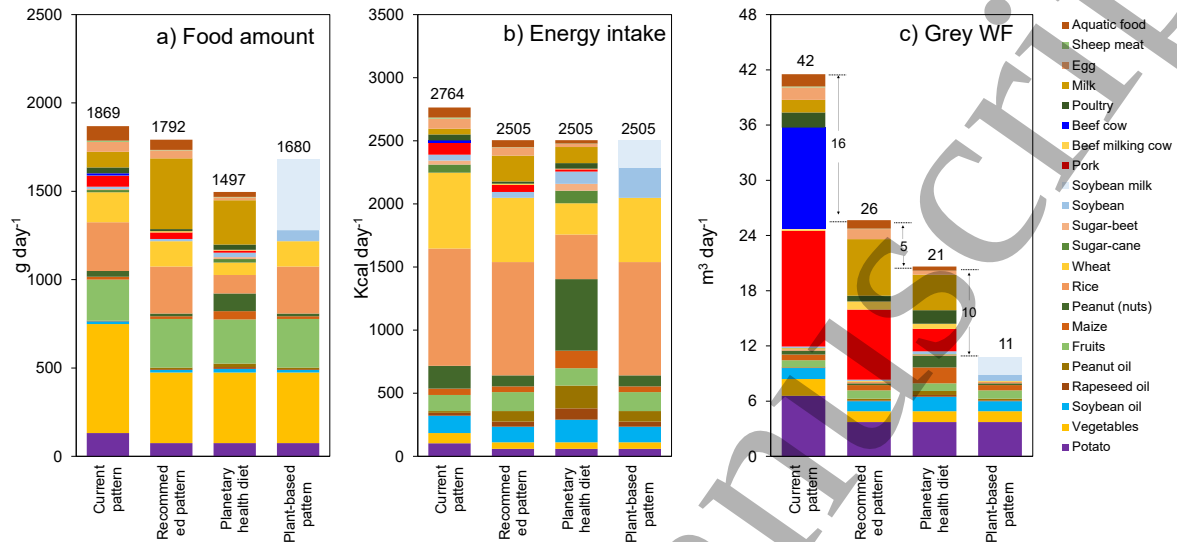


Figure 5 Food intake (a), energy intake (b) and grey water footprints (c) for four Chinese food consumption patterns: the current pattern based on China's food supply in 2020, the governmental recommended pattern based on China's daily food recommendation, the planetary health diet based and the proposed plant-based food consumption pattern in which animal foods and fish are replaced by soybean and potato foods based on the recommended diet. Numbers between bars in (c) indicate the absolute difference in grey WF (m³ day⁻¹) between adjacent dietary patterns (e.g., 16 indicates that the current pattern's grey WF is 16 m³ day⁻¹ larger than the recommended pattern).

Figure 5 shows that food choices that make up the four food consumption patterns have a huge effect on grey WFs. The current Chinese dietary pattern exhibits the largest grey WF of the four patterns. For the current pattern, the total daily food amount consumed (Figure 5a), energy intake (Figure 5b) and grey WF (Figure 5c) are largest with 1869 g day⁻¹, 2764 Kcal day⁻¹ and 42 m³ day⁻¹. The current pattern's grey WF is 61% larger than the recommended pattern (26 m³ day⁻¹), 100% larger than the planetary health diet (21 m³ day⁻¹), and 282% larger than the plant-based pattern (11 m³ day⁻¹) (Figure 5c, numbers between bars show the absolute differences in m³ day⁻¹). These differences reflect both the higher energy intake of the current pattern and its greater reliance on animal-based foods. When

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expressed per 1000 Kcal, the current pattern (15 m³ per 1000 Kcal) still exhibits the largest grey WF among all patterns, followed by the recommended pattern (10 m³ per 1000 Kcal), the planetary health diet (8 m³ per 1000 Kcal), and the plant-based pattern (4 m³ per 1000 Kcal) (estimated based on Figure 5b and 5c), confirming that dietary composition, not just total energy, drives the difference. Vegetables, fruits, rice and wheat contribute 70% to the total daily food amount consumed and 63% to the daily energy intake, but the contribution to the grey WFs is only 7%. The largest grey WF contributor is pork with 30%, followed by beef (27%) and potato (16%), although their contributions to the total food amount and energy intake are only 11% and 8%, respectively. This disproportionate contribution of animal products is the primary driver of grey WF differences across dietary patterns. The contribution of animal and aquatic foods to the total food amount and energy intake is small (18% and 13%), but it contributes 71% to the total grey WF.

The recommended diet shows a smaller daily food amount (1792 g day⁻¹), energy intake (2505 Kcal. day⁻¹) and grey WF (26 m³ day⁻¹) compared to the present pattern, decreasing food amounts by 4%, energy intake by 9% and grey WFs by 38%. When comparing the current and recommended pattern, milk consumption increases 4.5 times from 88 to 400 g day⁻¹, however, as a scenario, all beef is assumed to be produced by milking cows, resulting in 65% smaller grey WFs for beef. This assumption is made to demonstrate the potential benefit of beef as a co-product of dairy production, but it does not reflect the current Chinese beef supply mix, which is predominantly from specialized beef cows. Animal and aquatic foods still contribute 68% to the total daily grey WF in the recommended diet. A reduction of animal and aquatic food quantities and replacement of crop based food with more energy intensive crop based foods, like peanuts (nuts), in combination with additional sugar, turns the recommend pattern into the planetary health diet with a daily grey WF of 21 m³ day⁻¹. The contribution of animal and aquatic food is 45%. The plant-based pattern in which animal foods and fish are replaced by soybean products, like soybean burger and soybean milk, shows a total daily grey WF of 11 m³ day⁻¹, substantially smaller than the recommended food consumption pattern (-58%) and current pattern (-74%). These findings suggest that even partial shifts toward plant-based diets can gain substantial water pollution reductions.

Figure 6 shows the temporal changes of pesticide related grey WFs for the four Chinese food consumption patterns: a) the current food consumption pattern; b) the recommended pattern; c) the planetary health diet; and d) the plant-based food consumption pattern.

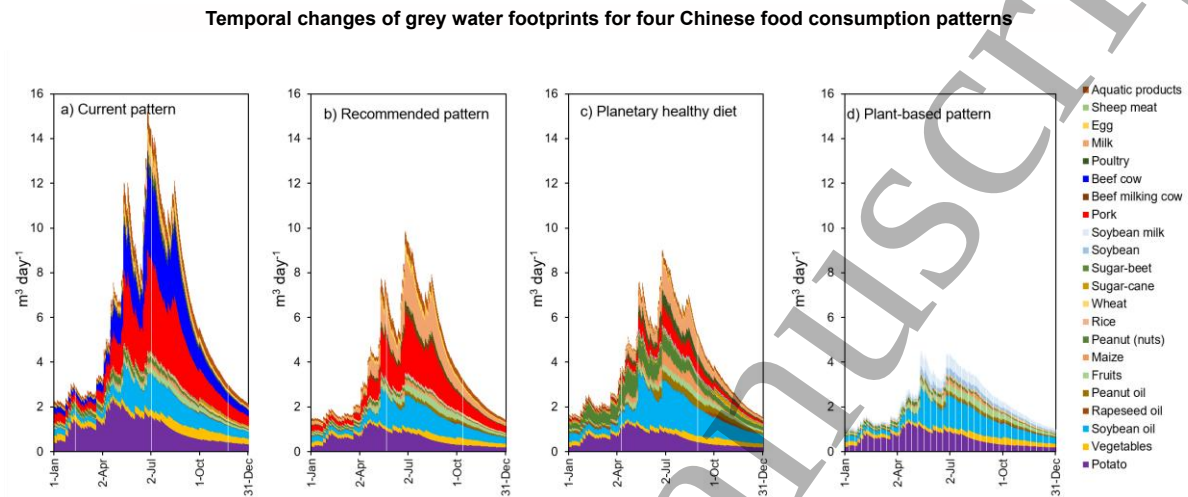


Figure 6 Temporal changes of pesticide related grey WFs for four Chinese daily food consumption patterns: a) the current food consumption pattern based on China's food supply in 2020; b) the recommended pattern based on China's governmental daily food recommendation, c) the planetary health diet; and d) the plant-based pattern in which animal foods and fish are replaced by soybean foods and potato.

Note: The values in Figure 6 are calculated using pesticide-specific degradation half-lives (following Yi et al., 2024b) and represent temporal pesticide-related grey WFs at specific days, whereas Figure 5c presents annual average total grey WFs (including all pollutants) under the conservative no-degradation assumption. These two sets of values are derived from different methodological approaches and are not directly comparable in magnitude.

Figure 6 shows that the temporal grey WFs for the four food consumption patterns all peak during the cropping season from June to August. Grey WFs exhibit pronounced seasonal peaks corresponding to the cropping season, when pesticide application for summer crops is concentrated. Total daily grey WFs depend on the food consumption pattern and decrease in the following order: current food consumption pattern ($15 \text{ m}^3 \text{ day}^{-1}$), recommended pattern ($10 \text{ m}^3 \text{ day}^{-1}$), planetary health diet ($9 \text{ m}^3 \text{ day}^{-1}$) and the plant-based pattern ($4.5 \text{ m}^3 \text{ day}^{-1}$) under the dynamic method (Figure 6), which incorporates pesticide

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application timing and degradation. Figure 6a and 6b also show that animal and aquatic foods dominate the grey WFs in summer, but the contribution in winter is relatively small. For example, the grey WFs of pork and beef are largest in summer when feed crops are grown with WFs of 4.5 and 3.8 m³ day⁻¹, contributing 55% to the temporal grey WF, and smallest in winter with grey WFs of 0.27 (pork) and 0.17 m³ day⁻¹ (beef) contributing 22% to the total temporal grey WFs. This seasonal pattern reflects the timing of feed crop production for animal feed and has practical implications for water quality management: targeted interventions during summer months, such as enhanced monitoring of pesticide runoff, could effectively reduce peak pollution loads.

5 Discussion

5.1 Study relevancy

This study establishes, for the first time, the consumption-to-production perspective for pesticide-related water pollution using a consumption-based grey water footprint (WF) framework using China as the case study area. The analysis builds on two earlier studies on water pollution from agriculture in China (Yi et al., 2024a, 2024b), incorporating information on food waste and losses in production chains (Xue et al., 2021) and data on four different food consumption patterns. The 18 pesticides were selected based on farmer surveys as the most frequently applied pesticides for these crops rather than on the share of total pesticide use, with the underlying data drawn from Zhang et al. (2015) and Yi et al. (2024a). We estimated that the present pesticides accounted for around 60% of the total use of pesticide in China (Jing et al., 2022). The 12 crops (maize, rice, wheat, soybean, potato, peanut, rapeseed, sugarcane, sugar beet, tea, cotton, and tobacco) and 2 crop categories (vegetables and fruits) together account for 97% of China's national crop planting area and represent 95% of total crop production. The 14 crop-based foods analyzed in Section 3 are covered by these 12 crops and 2 categories, with the exception of tea, cotton, and tobacco which are not food commodities and are therefore excluded from the dietary analysis. The four key findings are: (i) Pesticides—not nitrogen or phosphorus—dominate

grey WFs for most food commodities at the farm gate. This finding shifts the focus of water pollution mitigation from nutrient management alone to include pesticide reduction strategies. This implies that agricultural extension programs and water quality regulations should prioritize pesticide application practices, particularly for high-impact crops such as maize and potato. (ii) Supply chain losses substantially amplify grey WFs for most foods. For potatoes, the grey WF (m^3/kg) more than doubles from farm gate to consumer. This amplification effect suggests that food loss and waste reduction can serve as an effective water pollution mitigation strategy, especially for foods with high initial grey WFs. (iii) The current Chinese dietary pattern has the largest grey WF ($42 \text{ m}^3 \text{ day}^{-1}$), with animal products contributing 71% to the total grey WF despite accounting for only 18% of food intake. The government-recommended pattern reduces grey WF by 38%, the planetary health diet by 50%, and the plant-based pattern by 74% relative to the current pattern. These reductions demonstrate that even partial shifts toward plant-based diets contribute to substantial water pollution reductions. (iv) The seasonal analysis reveals that grey WFs peak during summer (June–August), driven primarily by feed crop production for pork and beef. Our temporal analysis (Figure 6), which incorporates pesticide application timing and degradation, reveals critical periods for pesticide management, the peaks in Figure 6 show when pesticide concentrations in water bodies are highest, providing actionable information for temporally targeted interventions. Second, the dynamic method identifies critical periods for pesticide management—the peaks in Figure 6 reveal when pesticide concentrations in water bodies are highest, providing actionable information for temporally targeted interventions, such as application timing restrictions or enhanced monitoring during high-runoff periods.

Our study focused on China as the case study area, but results might be relevant for other countries as well.

5.2 Methodological considerations and uncertainties

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3 485 Several methodological choices and data limitations warrant discussion regarding their potential
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11 488 *5.2.1 Data sources and representativeness*
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16 490 This study applied data sources for agricultural production from the main producing areas assuming
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18 491 they are representative for China as a whole. This was done to indicate the relation between
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20 492 consumption patterns and grey WFs. We compiled feed ingredient data from farmer surveys conducted
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22 493 in the largest feed producing region, i.e., Northern China (Lu et al., 2024; Wang et al., 2007; Yi et al.,
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24 494 2023), assuming these data represent national average practices. A limitation is the temporal mismatch
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26 495 between the feed composition data (primarily from Wang et al., 2007) and the 2020 consumption
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28 496 patterns analyzed. While feed efficiency in China has improved since 2007, we lack updated national
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30 497 feed composition data; this uncertainty should be addressed in future research. The study selected the
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32 498 typical modern animal production systems that dominate animal production. They use feed ingredients,
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34 499 like maize and soybean, mostly grown in Northern China. We assumed that the contribution from
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36 500 traditional systems is very small.
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42 502 *5.2.2 Manure-related nutrient pollution*
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47 504 Our grey WF calculations for animal products exclude nitrogen and phosphorus leakage from animal
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49 505 manure, which in China may be discharged directly to water bodies or improperly managed. This
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51 506 exclusion is consistent with the pesticide-focused scope of our study, as manure-related pollution falls
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53 507 under nutrient (nitrogen and phosphorus) rather than pesticide pollution. It likely leads to
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55 508 underestimation of nutrient-related grey WFs for animal products. We acknowledge this as a scoping
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57 509 limitation and recommend that future research integrate both pesticide and nutrient pollution pathways
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59 510 for a more comprehensive assessment of water pollution from livestock systems.
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512 5.2.3 Pesticide environmental fate

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514 Our calculations are based on conservative behavior for pesticides assuming that no degradation occurs
515 before they reach water bodies. This assumption is consistent with the grey WF accounting guidelines
516 (Franke et al., 2013). This approach provides a worst-case estimate of pesticide-related water pollution.
517 In reality, pesticide environmental behavior varies considerably: some pesticides (e.g., glyphosate) are
518 moderately mobile and degrade relatively quickly, while others (e.g., atrazine) are persistent and highly
519 mobile (Franke et al., 2013). More refined methods that incorporate pesticide-specific degradation rates
520 and soil adsorption coefficients would reduce absolute grey WF estimates. In our temporal analysis
521 (Figure 6), we already apply pesticide-specific degradation half-lives, which are embedded in the
522 dynamic grey WF calculations to provide a more realistic scenario for seasonal patterns. Regarding
523 water quality standards, our previous work (Yi et al., 2024a) examined the effect of alternative standards:
524 if the maximum permissible concentration (MPC) were applied instead of our chosen standard, the total
525 grey WF for crops would be fifteen times larger. This indicates that using a more stringent standard
526 would strengthen rather than weaken the conclusion that pesticides dominate water pollution from
527 Chinese food production systems. We acknowledge that we have not tested the opposite direction—a
528 less stringent drinking-water-type standard—as such standards are designed for human health rather
529 than aquatic ecosystem protection and are therefore less relevant for our assessment. However, we
530 recognize that a formal bounded sensitivity analysis across alternative standards would strengthen the
531 study; we have identified this as a priority for future research.

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533 5.2.4 Temporal dynamics of pesticide pollution

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535 A key methodological contribution is the incorporation of pesticide application timing and degradation
536 half-lives into grey WF calculations. Pesticide degradation characteristics and water quality standards

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both influence the magnitude and timing of grey WF peaks. For example, cypermethrin, with relatively low allowed concentrations in water, dominates the total grey WF in summer, but its contribution quickly drops due to its relatively short half-life (Yi et al., 2024b). The temporal changes of grey WFs show multiple seasonal peaks, which are dominated by different pesticides and the corresponding application time. The chemicals (e.g. fertilizers and pesticides) are applied at different times during crop growing periods (Yi et al., 2024b). Without considering chemical application times and degradation, the level of water pollution is probably overestimated. However, in those cases when application is larger than degradation we see another effect, i.e., pesticide concentrations in the environment might increase year by year. This would mean that a smaller grey WF as the outcome of the second method is only valid looking at one growing season. More importantly, the dynamic method reveals critical periods for pesticide management—the peaks in Figure 6 identify when pesticide concentrations in water bodies are highest, providing actionable information for temporally targeted interventions (e.g., restricting pesticide application during high-runoff periods). Temporal information is highly relevant for water quality management practices, as it identifies when pollution levels are highest at short time and which chemicals are most polluting with concentrations increasing at the long run.

5.2.5 Regional heterogeneity and international trade

The study focuses on the domestic consumption-production perspective, consistent with China's national food security strategy, which prioritizes domestic production for staple crops. This study adopts a national systems perspective to provide policy-relevant insights. While China's dietary patterns and agricultural practices vary regionally, the central government's unified policy framework, such as the National Food and Nutrition Development Outline 2025–2030 (AGFEP 2026), operates at the national scale. Moreover, ongoing urbanization and integrated national food supply chains are driving convergence in dietary patterns across regions (Huang et al., 2024). Our findings are intended to inform overarching policy directions, which can subsequently be adapted to regional contexts. Regarding

international agricultural trade, we exclude its influence because China's staple foods (rice and wheat) are mainly produced in China itself, with self-sufficiency rates exceeding 95% (FAO, 2024).

5.3 Comparison with earlier studies

Our study supports previous findings, suggesting that such dietary shifts can reduce land use (Kastner et al., 2012), greenhouse gas emissions (Tilman and Clark, 2014), and water pollution related to nitrogen, phosphorus (Mekonnen and Hoekstra, 2015, 2018), and pesticides (Yi et al., 2024ab). Consistent with the framework proposed by Duchin (2005) for sustainable food consumption patterns, our results confirm that plant-based dietary patterns have lower environmental impacts. More broadly, our study is in line with earlier research across multiple environmental dimensions, including GHG emissions (e.g., Guzmán et al., 2021), land and water use (e.g., Kastner et al. 2012), nitrogen and phosphorus pollution (e.g., Mekonnen and Hoekstra, 2015, 2018), and water footprints of agricultural products (e.g., Berger et al., 2025). Those studies consistently indicate that animal-based foods contribute disproportionately to total environmental pressure.

5.4 Policy implications

In China, rapid urbanization and economic growth have led to a shift towards higher consumption of especially animal foods (meat, eggs and milk products), deviating from the traditional Chinese diet based on vegetables, whole grains, and crop based proteins (Huang et al., 2024). Our study shows that the present Chinese food consumption pattern has the largest grey WF, suggesting the greatest potential impact on water pollution and that a shift towards consumption of less animal products, e.g. the planetary health diet or a plant-based diet, decreases pollution. Our findings suggest four policy priorities. First, dietary structure optimization. Chinese nutrition experts should advocate a more balanced diet aligned with the principles of the planetary health or plant-based diet (Willett et al., 2019), which emphasizes diverse options such as legumes, nuts, and whole grains while reducing meat and

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processed foods. The disproportionate contribution of pork, beef and potato to grey WFs (73% of total, with only 11% of food intake and 8% of energy) suggests that reducing meat consumption offers the greatest potential for water pollution mitigation. The plant-based pattern's 74% reduction relative to the current pattern, and the planetary health diet's 50% reduction, demonstrate that even partial shifts toward plant-based diets yield substantial environmental benefits. Second, agricultural production optimization. China's intensive agricultural practices, particularly in pork production, heavily rely on maize for feed, contributing significantly to pesticide pollution and other environmental effects. The dominance of pesticides in grey WFs—particularly for maize and potato (Figure 3)—indicates that improved pesticide management should be prioritized alongside nutrient reduction strategies. Better pesticide management, including less polluting pesticides, e.g. for maize or potato production, could also contribute to the decrease of grey WFs (Yi et al., 2024a). Third, supply chain loss reduction. The amplification effect documented in Section 4.2—where grey WFs increase by up to 117% for potatoes due to supply chain losses—highlights food loss and waste reduction as a complementary strategy. Fourth, policy coordination. The dynamic method (Figure 6) identifies seasonal peaks in grey WFs during June–August, revealing critical periods when pesticide concentrations in water bodies are highest. This provides an evidence base for temporally targeted interventions, such as restricting pesticide application during periods of high runoff risk, which could effectively mitigate peak pollution loads. A shift in consumption patterns towards more sustainable and health conscious choices could drive the optimization of agricultural production systems, reducing reliance on pesticide intensive crops and mitigating broader environmental emissions and pollutions. By promoting dietary patterns that are both nutritionally beneficial and environmentally sustainable, China can enhance food security, reduce its environmental burden, and contribute to global efforts in combating climate change. The results of our study showing that the planetary health diet and a plant-based diet cause less water pollution could be a reason for the Chinese government to adapt their nutritional guidelines. In general, the planetary health and plant-based food consumption patterns complying to nutritional requirements not only contribute to improved water quality but also to efforts to remain within the planetary boundaries for freshwater use, greenhouse gas emissions and land use.

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618 5.5 Future research directions

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620 Several directions for future research emerge from this study. First, while our analysis relies on
621 secondary data from farmer surveys and national statistics, future research could benefit from primary
622 data collection across multiple regions and production scales to improve accuracy. Second, our study
623 focused on pesticides, but excluded other pollutants such as antibiotics, which have emerged as a
624 significant environmental concern (Wang et al., 2024). Future studies should expand the pollutant scope
625 for a more comprehensive assessment. Third, we adopted a national systems perspective, but China
626 exhibits significant regional variation in dietary patterns and pesticide use intensity (Yi et al., 2024b).
627 Future research should incorporate regional heterogeneity to support localized implementation
628 strategies. Fourth, our grey WF calculations assumed conservative behavior for pesticides; future
629 studies could refine these estimates by incorporating pesticide-specific degradation rates. Fifth, we
630 excluded agricultural trade from our analysis, but China is a significant importer of feed crops, which
631 may indirectly contribute to water pollution in exporting countries. Future research should adopt a
632 global perspective to assess the full environmental footprint of Chinese food consumption.

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634 6 Conclusions

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636 This study made an assessment of the relation between food consumption patterns in China and water
637 pollution using the grey WF as an indicator. By establishing the consumption-to-production perspective
638 for pesticide-related water pollution, we provide quantitative evidence to support coordinated policy
639 interventions across the food supply chain. Pesticides, in comparison to N and P, dominate grey WFs
640 for most foods. However, there are significant variations of grey WFs within and among different food
641 commodity categories. Meanwhile, the grey WFs increase from the production stage to the consumer,

642 due to losses in the supply chain. All of them contribute to water pollution related to pesticides for final
643 food consumption for different consumption patterns.

644 In China, the current food consumption pattern has the largest food availability, nutritional energy
645 supply and grey WF ($42 \text{ m}^3 \text{ day}^{-1}$) in comparison with the Chinese governmental recommended pattern
646 ($26 \text{ m}^3 \text{ day}^{-1}$), the planetary health diet ($21 \text{ m}^3 \text{ day}^{-1}$) and the proposed plant-based pattern ($11 \text{ m}^3 \text{ day}^{-1}$). The current pattern's grey WF is 61% larger than the recommended pattern, 100% larger than the
647 planetary health diet, and 282% larger than the plant-based pattern. For the current pattern, vegetables,
648 fruits, rice, and wheat make up 70% of the total daily food consumed and 63% of daily nutritional
649 energy supply, however, they only contribute 7% to grey WFs, whereas pork, beef, and potato which
650 account for just 11% of food intake and 8% of energy, contribute 30%, 27% and 16% respectively.
651 Animal and aquatic foods, making up 18% of total food intake and 13% of nutritional energy, contribute
652 71% to the total grey WF.

654 In the recommended diet, milk consumption increases 4.5 times, and the shifting beef production
655 from beef cows to milking cows reduces the grey WF for beef by 65%. Despite this, animal and aquatic
656 foods still account for 68% of the total grey WF. Adopting a planetary health diet reduces the daily grey
657 WF, with animal and aquatic foods contributing 45%, while a plant-based diet further lowers it to $11 \text{ m}^3 \text{ day}^{-1}$, significantly less than both the recommended (-58%) and current patterns (-74%). Meanwhile,
658 the temporal grey WFs peak during the cropping season, with the current food consumption pattern
659 exhibiting the highest grey WF peaks and the plant-based pattern showing the lowest. Comparatively,
660 animal and aquatic foods significantly increase the grey WFs during summer, particularly in the current
661 and recommended patterns, while their impact is much smaller in the planetary health diet and plant-
662 based pattern. This seasonal pattern has practical implications for water quality management: targeted
663 interventions during summer months could effectively reduce peak pollution loads.

665 Our analysis highlights that present Chinese food consumption patterns substantially affect grey WFs,
666 with animal foods and potato consumption, and related pesticide use for agricultural production,
667 contributing significantly to the grey WF, an indicator of potential water pollution. These findings are
668 in line with earlier studies indicating the impact of consumption patterns with much animal foods on
669 greenhouse gas emissions, energy use, water and land footprints. Transitioning to more sustainable food

consumption patterns with less animal foods and better pesticide management for especially maize and potatoes not only reduces GHG emissions and natural resource use, but also reduces water pollution, emphasizing the importance of adopting strategies to make food consumption patterns more sustainable in an effort to mitigate environmental effects, not only in China, but on a regional and even global scale. The study shows that the planetary health and plant-based food consumption patterns, complying to nutritional requirements, contribute most to improved water quality. Although the study has taken China as the case study area, our methodology is also relevant on a global scale, where food consumption choices affect environmental sustainability, including water pollution with country-specific data provided.

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Credit Author Statement

Conceptualization, J. Yi, P.W. Gerbens-Leenes; Data collection and analysis, J. Yi; Writing original draft preparation, J. Yi; Editing and Revising, P.W. Gerbens-Leenes and L. Wang; Contribution to study design, P.W. Gerbens-Leenes; Visualization, J. Yi, P.W. Gerbens-Leenes and L. Wang.

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